Mamba-MPSE: Multipattern State Evolution Based on the Mamba Model for Intraclass Heterogeneous Wetland Classification With UAV Hyperspectral Imagery

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***Abstract*—In hyperspectral imagery, intraclass heterogeneity often arises from local environmental variations and spatial distribution nonuniformity, leading to significant discrepancies across spectral and spatial dimensions and limiting the capacity of unified modeling mechanisms. The Mamba model, built upon the selective state-space model (SSM), integrates local perception with long-range dependency modeling; however, its state evolution matrix adopts a single evolution pattern, lacking dynamic adaptability to heterogeneous features. To address this limitation, we propose a unified framework that integrates multipattern state evolution (MPSE) and dual-path spectral–spatial (DP-SS) modeling. MPSE introduces multiple learnable evolution patterns for hidden states, enabling each sample to dynamically integrate diverse historical information and thereby capture intraclass spectral–spatial heterogeneity. Meanwhile, DP-SS models feature along both spectral and spatial paths, enhance the representation of spectral trends and spatial structures, and fuse the two paths for adaptive global feature integration. The synergy of these modules strengthens the model's ability to discriminate intraclass heterogeneous land covers. Experiments conducted on a self-constructed UAV hyperspectral dataset of the Yellow River Wetland demonstrate that the proposed method significantly outperforms existing approaches, achieving an overall accuracy (OA) of 98.27%, an average accuracy (AA) of 98.84%, and a Kappa coefficient of 98.63%, offering a new perspective for efficient and fine-grained hyperspectral image classification.**

***Index Terms*—Deep learning, hyperspectral imagery, Mamba, sequence modeling, UAV remote sensing classification.**

I. Introduction

HYPERSPECTRAL data, with its richer spectral band information, offers unprecedented potential for achieving high-precision remote sensing classification. However, while the increased number of bands provides more detailed information for object identification, it also introduces significant challenges in processing and modeling. Recent research has shown that optimizing spectral band selection can improve classification performance by reducing redundancy and enhancing computational efficiency [1]. This highlights a major challenge for modeling methods: how to effectively handle the rich information inherent in high-dimensional hyperspectral data while maintaining efficient processing [2], [3], [4], [5], [6]. To address these challenges, various modeling methods have been proposed in recent years. Transformer architectures have demonstrated remarkable performance in long-sequence modeling through self-attention mechanisms; however, their quadratic computational complexity limits their scalability to high-dimensional hyperspectral features [7], [8], [9], [10]. Convolutional neural networks (CNNs), while computationally efficient, are restricted by local receptive fields, making it difficult to capture long-range dependencies and global features. As a result, they often struggle to accurately identify homogeneous land covers distributed across spatial regions in wetlands [11], [12], [13]. As a selective state-space model (SSM), Mamba achieves unified modeling of local details and long-range dependencies with linear complexity, showing strong scalability and efficiency in sequence learning. Specifically, it operates by decomposing the input sequence into a set of latent states, which are iteratively updated through a selective transition mechanism that emphasizes informative positions while suppressing redundant ones. This design enables Mamba to efficiently capture both fine-grained local patterns and global contextual dependencies, providing a promising pathway for modeling high-dimensional heterogeneous hyperspectral sequences [14], [15], [16].

In recent years, several studies have attempted to incorporate the Mamba model into hyperspectral remote sensing and develop customized extensions. Spatial–spectral Mamba for hyperspectral image classification (MambaHSI) was the first

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to apply Mamba to hyperspectral tasks, designing spectral Mamba (SpeMB) and spatial Mamba (SpaMB) modules to model band-wise dependencies and pixel-level spatial structures, respectively, with a fusion module for spectral–spatial feature integration; however, it still relies on fixed-scale modeling and cannot adapt flexibly to dynamic spectral variations [17]. Subsequently, 3-D-spectral–SpaMB for hyperspectral image classification (3DSS-Mamba) constructs voxel-level spectral–spatial interaction paths via 3-D selective scanning, enhancing spatial structure modeling, yet the spectral evolution patterns remain uncharacterized at the mechanism level [18]. Furthermore, spectral–spatial adaptive Mamba for hyperspectral image classification (HyperMamba) introduces spectral enhancement and neighborhood-adaptive scanning, leveraging spatial neighborhood information to optimize spectral modeling, achieving superior performance in locally complex regions, but lacks a unified global modeling mechanism, limiting its ability to capture subtle cross-region spectral variations [19]. Some works have attempted to integrate Mamba with transformer architectures; for instance, integrating Mamba and transformer experts for hyperspectral image classification (HSI-MFormer) employs multiscale token grouping for joint modeling, enabling cross-scale feature integration, yet interactions between paths remain shallow [20]. Recently, multidirectional state propagation for efficient hyperspectral image classification (MambaHSI+) adopts bidirectional state propagation and spectral trajectory modeling, improving spectral continuity and balancing accuracy with efficiency, but still primarily relies on global unified modeling and lacks capabilities for recognizing spectral–spatial heterogeneity [21].

The issue of "intraclass heterogeneity" in hyperspectral remote sensing, particularly in wetland regions, is often driven by natural geographic variations, resulting in significant discrepancies in spectral curves and textures among homogeneous land covers across different regions [22]. For instance, vegetation with high and low water content exhibits distinct near-infrared responses, and densely versus sparsely distributed areas show markedly different texture structures; as spatial distance increases, these differences become even more pronounced. This intraclass heterogeneity makes it particularly challenging to accurately identify homogeneous land covers in wetland remote sensing imagery [23], [24]. The Mamba model is well-suited for hyperspectral remote sensing tasks due to its selective SSM framework [25], [26], [27]. This framework controls the evolution of hidden states through the state evolution matrix, enabling dynamic updates of historical information and facilitating unified global modeling. Such characteristics allow Mamba to handle high-dimensional data efficiently, balancing local details with long-range dependencies, and offering linear complexity and strong scalability. These advantages make Mamba an ideal base model for addressing the issue of intraclass heterogeneity. While existing Mamba variants have explored various directions, they generally rely on fixed-scale modeling or structural stacking, which limits their flexibility in adapting to dynamically changing spectral and spatial features. These approaches fail to capture the subtle variations of spectral and spatial heterogeneity, especially when dealing with complex cross-region discrepancies, thus limiting their effectiveness in addressing the intraclass heterogeneity problem in wetland hyperspectral imagery.

To address these challenges, we propose a unified modeling framework targeting intraclass heterogeneity, integrating multipattern state evolution (MPSE) with a dual-path spectral–spatial (DP-SS) structure based on the Mamba modeling mechanism. The core idea of MPSE is to introduce multiple evolution patterns for hidden states, allowing homogeneous land covers to dynamically integrate historical information in a differentiated manner during training. Specifically, each pattern corresponds to a learnable evolution rule that adaptively modulates the contribution of historical information according to spatial or spectral variations within the same class, thereby preserving class consistency while capturing intraclass heterogeneity. Through MPSE, the model effectively addresses pronounced spectral and spatial differences among samples, alleviating the limitations of a single evolution pattern. Building on this, DP-SS models feature along spectral and spatial paths: the spectral path enhances consistent modeling of dominant spectral trends, while the spatial path improves robustness to spatial perturbations. Furthermore, two sample-level adaptive mechanisms are introduced: sample-level differential modulation (SDM) assigns independent dynamic evolution strengths to each sample, preventing information coupling across classes within a batch; sample-level conditional fusion (SCF) incorporates adaptive weights during path fusion, enabling the model to dynamically balance the contributions of spectral and spatial features for more targeted joint representation.

The main contributions of this work are summarized as follows.

1) A self-constructed UAV hyperspectral dataset of the Yellow River Wetland, supported by field surveys and land cover annotations, highlighting intraclass heterogeneity in both spectral and spatial dimensions. This dataset provides a reference for data-problem-model co-innovation in hyperspectral classification.

2) An MPSE mechanism that introduces multiple learnable dynamic patterns for hidden states. This enables dynamic modeling of spectral and spatial variations among samples of the same class, enhancing the capture of heterogeneous features.

3) A differential processing spectral–spatial (DP-SS) framework designed to improve perception of dominant spectral trends and spatial perturbations while enabling interpath feature interaction. SDM and SCF provide each sample with independent dynamic evolution trajectories and fusion weights, achieving adaptive regulation and coordinated spectral–spatial representation.

II. Data Construction and Problem Analysis

A. UAV-Based Hyperspectral Remote Sensing Data Acquisition System

The UAV hyperspectral remote sensing data acquisition system used in this study consists of a UAV platform and an onboard hyperspectral imaging sensor, as shown in Fig. 1. The UAV platform adopts a multirotor design, capable of both

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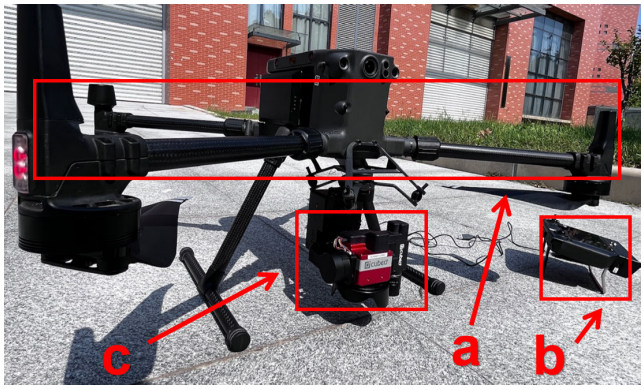


Fig. 1. UAV-based hyperspectral remote sensing data acquisition system. (a) UAV platform, (b) control handle, and (c) hyperspectral gimbal.

TABLE I
MAIN SPECIFICATIONS OF THE UAV PLATFORM

Categories	Parameter
Dimensions	810 × 670 × 430 mm (L × W × H)
Wheelbase	895 mm
Max Takeoff Weight	9 kg
Max Payload	2.7 kg
RTK Position Accuracy	1 cm + 1 ppm (horizontal); 1.5 cm + 1 ppm (vertical)
Max Horizontal Speed	23 m/s (S-mode); 17 m/s (P-mode)
Max Flight Altitude	5000 m-7000 m
Max Wind Resistance	12 m/s
Max Flight Time	55 min

hovering and executing planned flight paths. The hyperspectral imaging sensor is a dedicated airborne device mounted on a two-axis gimbal, enabling image stabilization and vibration reduction during data acquisition.

1) UAV Platform: The UAV platform employed in this experiment is the Matrice 300 RTK multirotor drone developed by DJI Innovations, Shenzhen, China. The UAV system comprises four main components: power system, flight control system, avionics system, and airframe structure. The main technical specifications of the UAV are listed in Table I.

2) Airborne HSI Imager: The onboard hyperspectral sensor used in this study is the X20P hyperspectral imaging device manufactured by Cubert GmbH. The X20P is based on light-field imaging technology and features a 20MP ultrahigh-definition CMOS sensor. It adopts a snapshot imaging approach to rapidly capture hyperspectral images with more than 160 spectral bands, continuously covering the wavelength range from 350 to 1000 nm. With dual GigE camera interfaces, it supports an image frame rate of up to 5 Hz (1886 × 1886 pixels/frame). The technical specifications of the hyperspectral sensor are shown in Table II.

3) Data Preprocessing: During data processing, white reference calibration was applied to convert raw digital numbers to reflectance, compensating for illumination variations and sensor spectral response. Dark current correction was also performed to remove baseline noise inherent to the sensor. Furthermore, ultraviolet bands (350–380 nm) with high noise and low information quality were excluded to enhance the reliability of the spectral data and improve subsequent analyses.

TABLE II
MAIN SPECIFICATIONS OF THE X20P AIRBORNE HSI IMAGER

Categories	Parameter
Dimensions	169 mm × 145 mm × 172 mm
Structural Weight	Integrated gimbal design, total weight $<$ 1.5 kg
Spectral Range	350-1000 nm
Spectral Resolution	4 nm
Number of Bands	164
Hyperspectral Resolution	1886 × 1886 pixels/frame
Hyperspectral Imaging Speed	$>$ 2 Cubes/s (1886 × 1886 pixels per cube)
Detector Size	12.16 × 12.16 mm
Focal Length	19 mm
Optical Array / Field of View	66 elements / 35°

B. Study Area

In this study, we selected the Yellow River Wetland Park, located in Huiji District, Zhengzhou City, Henan Province, as the research area. This region lies within the Yellow River Basin, near the coordinates of longitude 113°39′37.38″, E and latitude 34°55′3.97″, N, and represents a typical wetland ecosystem. The Yellow River Wetland Park was officially designated as a National Wetland Park in 2015. To facilitate hyperspectral remote sensing classification, a representative sample plot characterized by complex land cover types and diverse vegetation was selected within the park, where field surveys and vegetation classification were conducted, as shown in Fig. 2. The main land cover types are illustrated in Fig. 3.

C. Presentation of Our Constructed Dataset

The UAV-borne hyperspectral dataset of the Yellow River Wetland constructed in this study includes data of the same scene captured in two seasons (autumn and winter). Both datasets correspond to the same study area, with only differences in the data collection seasons. Based on field survey records, 8 representative land cover types were selected as classification targets; for the hyperspectral images of autumn and winter, experts manually outlined the regions of interest (ROIs) and assigned category labels according to the field survey results to ensure the accuracy and consistency of the annotations for both datasets, and the pseudo-color images and corresponding ROI annotation results of the autumn and winter data are shown in Fig. 4.

A unified random sampling strategy was adopted for dataset division: 5% of the annotated samples from both datasets were used for training, with the remaining 95% reserved for testing. This design simulates real-world scenarios with limited training samples, enabling a more accurate evaluation of the model's generalization ability. The dataset (incorporating autumn and winter data) encompasses eight typical wetland land cover categories, with all sample annotations verified through field surveys. The annotated pixel counts for different

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(a) (b)

Fig. 2. Research area. (a) Satellite image of the Yellow River Wetland Park in Zhengzhou. (b) Selected representative sample plot, where the UAV-acquired false-color hyperspectral data are displayed with bands corresponding to R (638 nm), G (550 nm), and B (470 nm).



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 3. Field-surveyed examples of land cover categories. (a) Willow, (b) Reed, (c) Poplar, (d) Cogon grass, (e) Miscanthus, (f) road, (g) Yellow River, and (h) Mudflat.

categories are identical across both datasets, exhibiting broadly consistent distribution patterns as shown in Table III. For instance, the Yellow River and willow categories constitute 1.05% and 0.85% of pixels in their respective images, while road and mudflat categories account for 0.29% and 0.42%, respectively. This reflects the naturally occurring category distribution imbalance within the wetland environment, providing a foundation for analyzing model performance across specific categories. In this study, autumn data serves as the primary UAV dataset for research, with winter data acting as auxiliary data; subsequent sections will introduce a corresponding primary and auxiliary dataset from public satellite imagery to establish a unified research framework.

TABLE III

SAMPLE NUMBERS OF TRAINING AND TESTING SETS ON THE YELLOW RIVER WETLAND (AUTUMN AND WINTER) DATASET

Hyperspectral Data				
Class No	Class Name	Train	Test	Percentage
1	Willow	46	880	0.85%
2	Reed	40	766	0.74%
3	Poplar	42	796	0.77%
4	Cogon Grass	34	663	0.64%
5	Miscanthus	44	838	0.81%
6	Road	16	300	0.29%
7	Yellow River	57	1087	1.05%
8	Mudflat	22	436	0.42%
Total		**291**	**5991**	**4.63%**

D. Problem Analysis and Modeling Motivation

This section analyzes the self-constructed UAV hyperspectral dataset of the Yellow River Wetland (autumn), focusing on the spectral and spatial heterogeneity characteristics within the same land cover classes to provide theoretical support for model design. The analysis covers four aspects: 1) spec-

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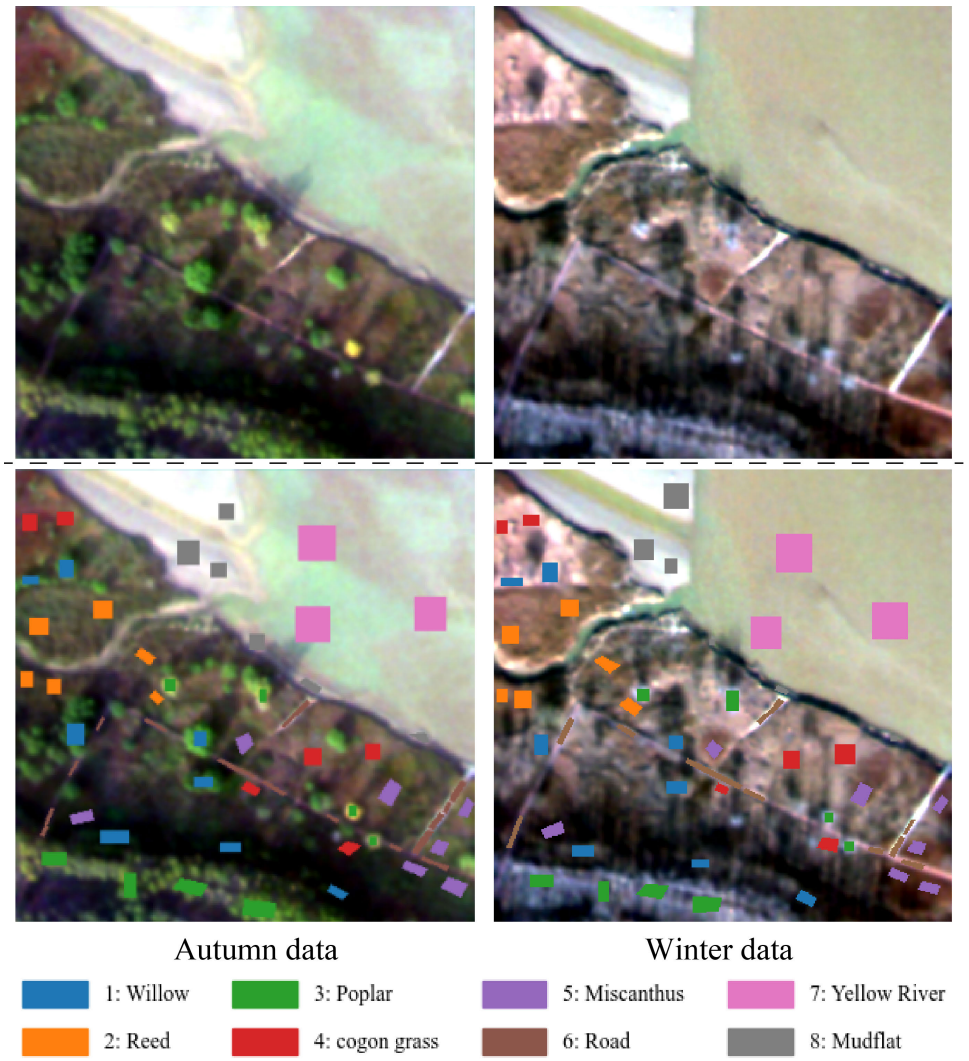


Fig. 4. False-color hyperspectral data and corresponding ROI annotations for autumn and winter (left: autumn; right: winter). The data were captured by a UAV at a flight altitude of 400 m. The upper row shows the false-color hyperspectral images with dimensions of $(330, 330, 150)$ (height $\times$ width $\times$ bands), with data sizes of approximately 62.5 MB for the autumn data and 61.8 MB for the winter data. The displayed bands correspond to R (638 nm), G (550 nm), and B (470 nm), and the data are stored in floating-point format. The lower row presents the corresponding ROI annotation maps, where different label colors indicate different land cover types as shown in the legend.

tral heterogeneity within the same class; 2) spatial texture heterogeneity within single bands; 3) spatial distribution discreteness; and 4) overall intraclass fluctuation and interclass trend differences.

1) Spectral Heterogeneity Within the Same Class: As shown in Fig. 5, the spectral curves of patches belonging to the same class exhibit high overall consistency in global trends, while significant variations occur in specific local bands. This "globally consistent yet locally heterogeneous" pattern reveals complex, fine-grained variation structures along the spectral dimension [28]. When relying solely on a single state evolution pattern, the model struggles to adequately capture such spectral differences, exposing its representational limitations. To address this, we propose an MPSE mechanism to enhance the model's capability in dynamically capturing fine-grained intraclass heterogeneity along the spectral dimension.

2) Spatial Texture Heterogeneity Within Single Bands: As shown in Fig. 6, in a single spectral band (Band 10), the extracted 9×9 spatial patches exhibit highly unstable and heterogeneous local textures, even within the same class, leading to limited discriminative capability of texture features [29]. This observation echoes the findings from the spectral dimension in Fig. 5: intraclass heterogeneity is not confined to a single dimension but represents a common challenge across both spectral and spatial domains. Therefore, the model must possess multipattern modeling capabilities in both dimensions simultaneously. Serving as the structural backbone, DP-SS decouples the spectral and spatial pathways for separate processing, while the core MPSE mechanism operates in parallel across both paths, ensuring that the model dynamically captures intraclass heterogeneity across multiple dimensions.

3) Spatial Distribution Discreteness: As shown in Fig. 7, different land cover classes exhibit distinct spatial distribution patterns in the 2-D plane: some classes (e.g., Yellow

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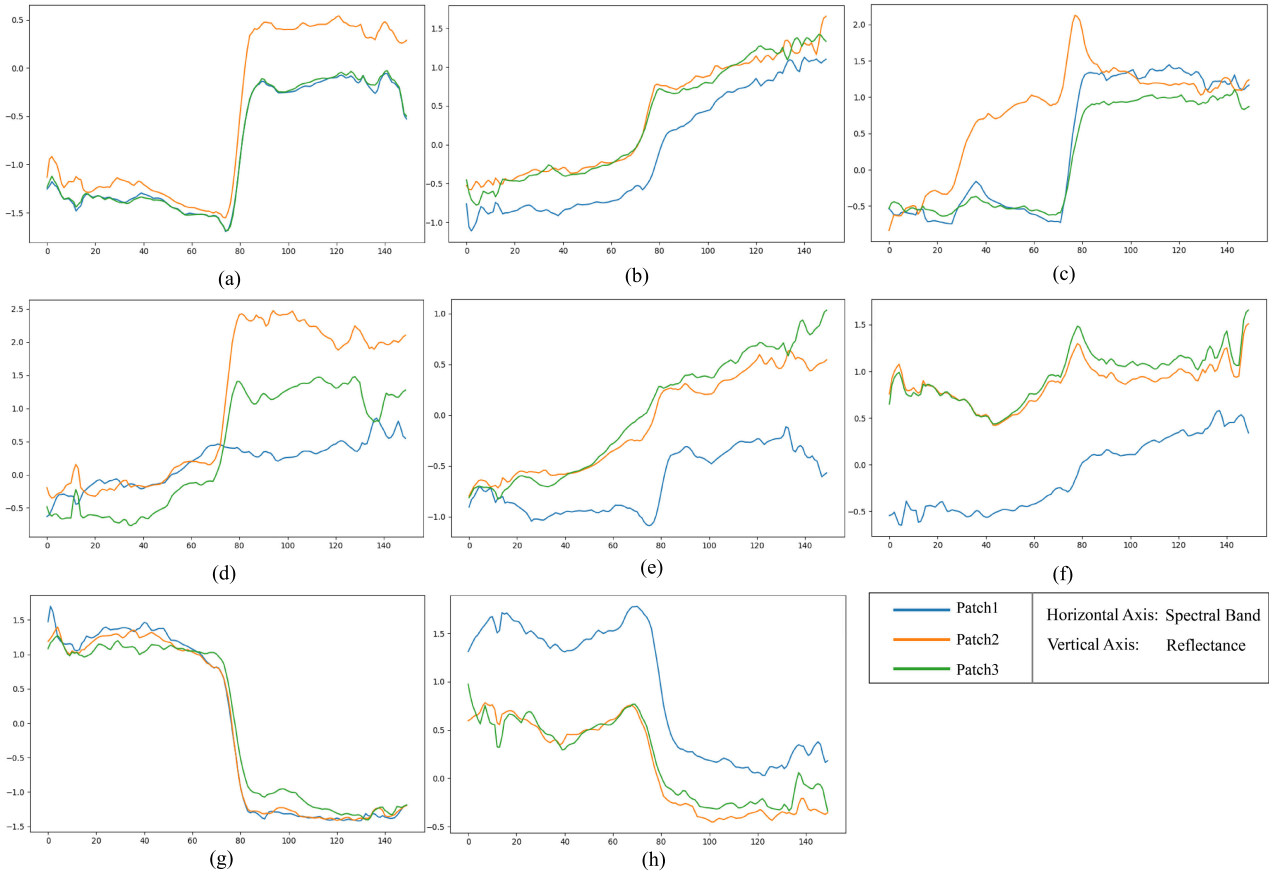


Fig. 5. This figure group presents the standardized spectral curves of each land cover class (a total of eight classes). For each class, three randomly selected spatial patches are shown, with reflectance variations plotted across 150 spectral bands (x-axis: 0–150 corresponding to the band range; y-axis: reflectance). Spectral curves were standardized using z-score normalization (subtracting the mean and dividing by the standard deviation of each spectral band across all pixels), highlighting intraclass spectral heterogeneity. (a) Willow, (b) Reed, (c) Poplar, (d) Cogon grass, (e) Miscanthus, (f) road, (g) Yellow River, and (h) Mudflat.

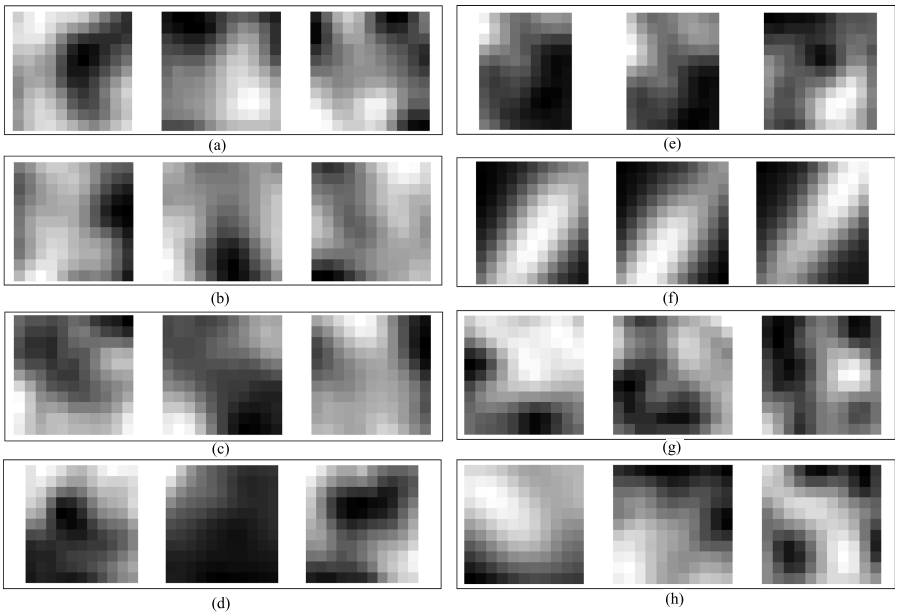


Fig. 6. This figure group shows the grayscale visualization of 11×11 spatial patches extracted around three randomly selected training pixels for each of the eight classes at spectral Band 10. Each subfigure corresponds to a single patch from one class. (a) Willow, (b) Reed, (c) Poplar, (d) Cogon grass, (e) Miscanthus, (f) road, (g) Yellow River, and (h) Mudflat.

River) are continuously clustered, while others (e.g., road) appear sparse, scattered, or even fragmented across regions. This spatial nonhomogeneity caused by long-distance dispersion is essentially another manifestation of intraclass

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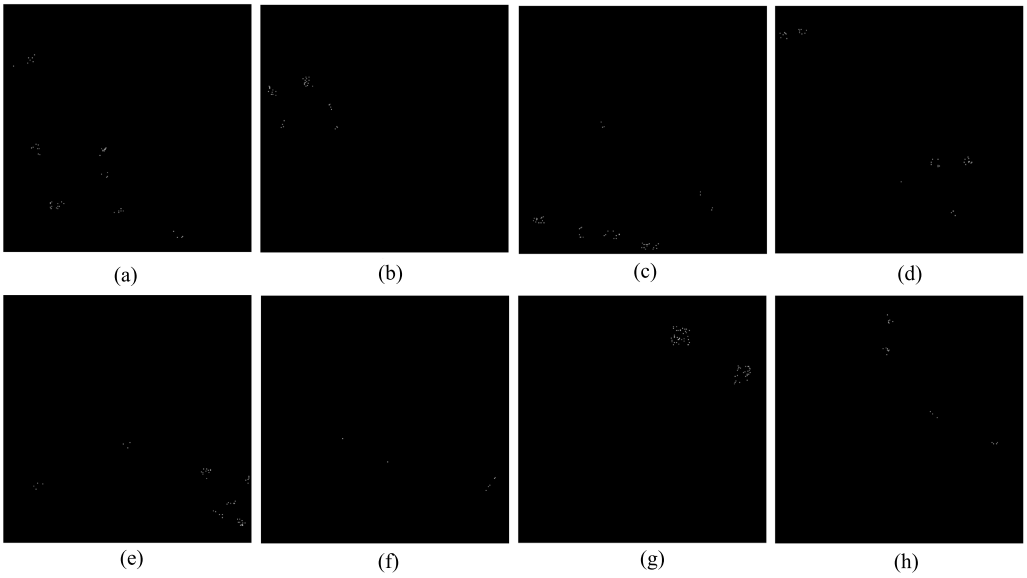


Fig. 7. This figure group presents the spatial masks of training samples for each of the eight classes. In each mask, the black background represents empty space, while clusters of white points indicate the spatial distribution of the corresponding land cover class. (a) Willow, (b) Reed, (c) Poplar, (d) Cogon grass, (e) Miscanthus, (f) road, (g) Yellow River, and (h) Mudflat.

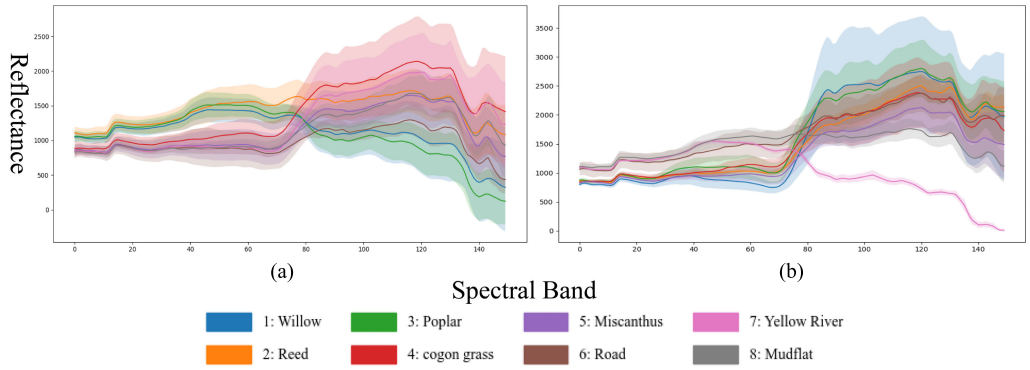


Fig. 8. This figure group shows the spectral reflectance curves of different land cover classes, with the x-axis representing spectral bands and the y-axis representing reflectance. The shaded areas around the curves, in matching colors, indicate the range of reflectance values for intraclass samples across bands. (a) Mean spectral curves for each class and (b) mean spectral curves with standard deviation for each class. The bottom legend indicates the corresponding land cover types represented by specific label colors.

heterogeneity [30]. When samples of the same class cannot rely on a unified spatial neighborhood for modeling, a single state evolution model struggles to balance local consistency with cross-region variation.

Combined with the observations from Figs. 5 and 6, a complete chain of multidimensional intraclass heterogeneity emerges—from local spectral discrepancies to unstable local spatial textures and finally to global spatial nonhomogeneity. Under this context, the MPSE mechanism is naturally motivated to operate in both spectral and spatial pathways, while DP-SS serves as the decompositional backbone that enables the model to dynamically adapt to intraclass heterogeneity across different scales.

4) Overall Intraclass Fluctuation and Interclass Trend Differences: As shown in Fig. 8, the mean spectral curves and standard deviations of each class further reveal both commonalities and variations along the spectral dimension: interclass overall trend differences provide a basis for discrimination, while intraclass band fluctuations reflect significant intraclass heterogeneity. This indicates that, while enhancing interclass separability, the model must also possess tolerance and adaptive capability for intraclass spectral variations to avoid over-generalization that could impair performance. Combined with the previously observed local and global spatial patterns, this spectral analysis further validates the central role of MPSE in dynamically capturing multipattern spectral heterogeneity, as well as the necessity of DP-SS in decoupling spectral and spatial pathways to support MPSE.

From the above multilevel analysis of spectral and spatial dimensions, it is evident that intraclass land cover exhibits both fine-grained differences and macroscale spatial nonhomogeneity [31], [32], [33]. These multidimensional heterogeneous characteristics not only confirm the ubiquity of intraclass heterogeneity in UAV hyperspectral wetland data but also provide direct guidance for model design. The phenomena revealed in this section constitute the foundation for the design of

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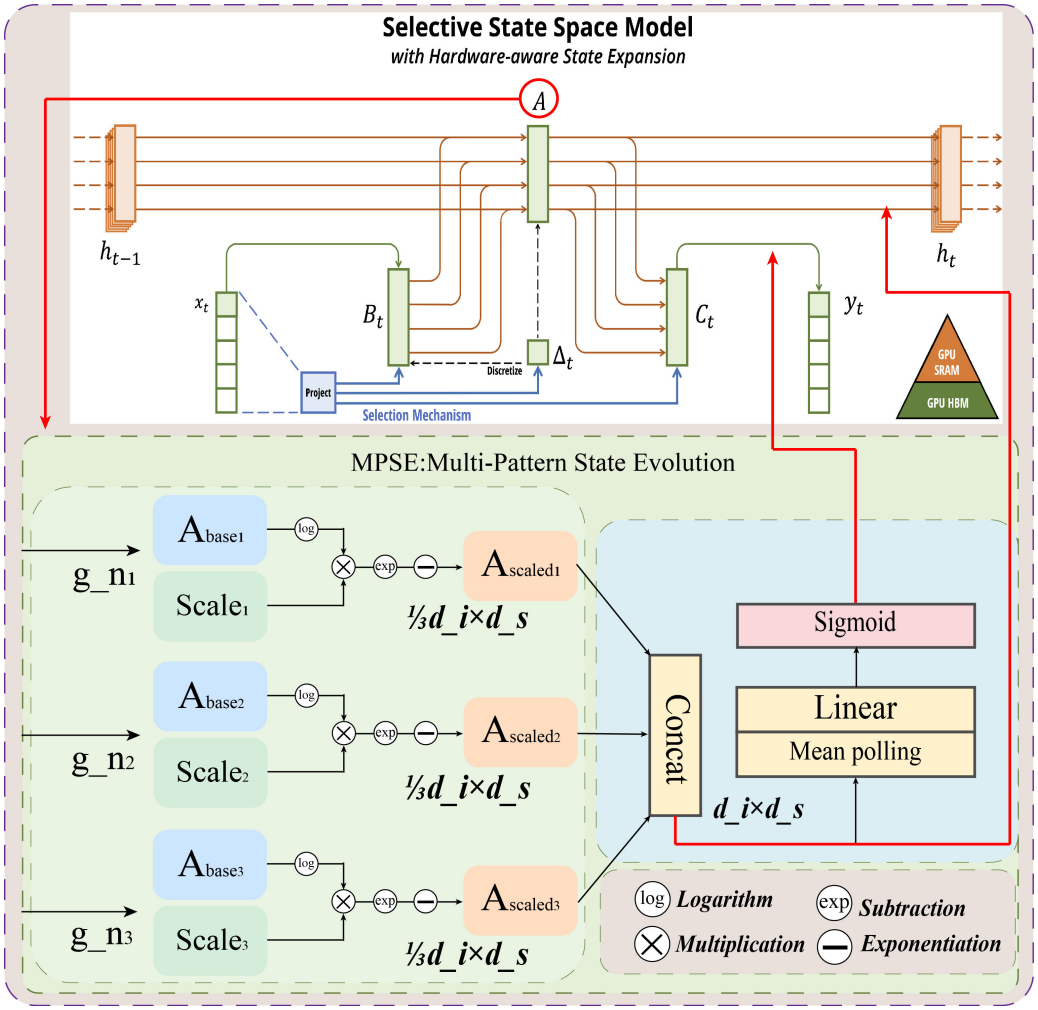


Fig. 9. Role of MPSE in the selective SSM. The upper part with a white background shows the original SSM structure, while the lower part depicts the three-mode state evolution. The outputs are processed through logarithm, learnable scaling factors, concatenation, and normalization to produce the weight vector. The model contains two streams: one is the core output that flows through the original SSM to update the hidden state h_t, and the other participates in computing the output y_t.

the MPSE and DP-SS dual-path structure, offering theoretical insights and practical References for subsequent modeling of intraclass heterogeneity in hyperspectral imagery.

III. MPSE AND DP-SS

A. Multipattern State Evolution

Motivated by the multilevel intraclass heterogeneity analysis presented in Section II-D, we observe that spectral–spatial variations within the same land cover class manifest in diverse forms and at different granularities. Specifically, locally fluctuating spectral responses, unstable spatial textures within single bands, and large-scale spatial nonhomogeneity across distant regions impose distinct requirements on feature state evolution behaviors. A single, fixed state evolution pattern is therefore insufficient to simultaneously accommodate these heterogeneous dynamics. To address this challenge, we introduce an MPSE mechanism, which employs multiple parallel state evolution patterns to accommodate heterogeneous feature dynamics exhibiting different variation behaviors. By enabling the model to adaptively select and combine multiple evolution modes, MPSE provides a flexible and effective means to capture complex intraclass spectral–spatial heterogeneity, thereby establishing an explicit and well-grounded connection between the model design and the multilevel empirical observations revealed in Section II-D.

In traditional state-space modeling, the state evolution matrix A is typically a single negative exponential decay matrix, used to describe the dynamic change of features along the temporal dimension. For example,

$$A = -\exp\left(\log\left([1, 2, \ldots, d_{\text{state}}]\right)\right) \tag{1}$$

where $A \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{state}}}$ is the state evolution matrix; d_{inner} is the hidden state dimension; and d_{state} is the state sequence length. This study proposes an MPSE framework, illustrated in Fig. 9, constructed with three groups of mode state evolutions.

1) Multipattern Exponential Decay Base Matrix: First, we define a multipattern base matrix A_{base} to represent multiple fundamental state evolution patterns. The state vector $t \in \{1, 2, \ldots, d_s\}$ is mapped to multiple patterns, each

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corresponding to a group

$$A_{\text{base}}^{(i)} = \log\left([1, 2, \ldots, d_s]\right), \quad i = 1, 2, \ldots, G \tag{2}$$

where G is the number of evolution patterns (groups); d_s is the state length; $A_{\text{base}}^{(i)} \in \mathbb{R}^{d_g \times d_s}$, and $d_g = d_{\text{inner}}/G$ is the hidden dimension slice for each pattern. The $\log(\cdot)$ function ensures numerical stability during the exponential mapping.

2) Learnable Scaling Factor: To allow each pattern to have a different decay rate, a learnable scaling factor α_i is introduced for each pattern

$$A_{\text{scaled}}^{(i)} = -\exp\left(\alpha_i \cdot A_{\text{base}}^{(i)}\right) \tag{3}$$

where $\alpha_i \in \mathbb{R}$ is the scaling factor for the ith pattern, automatically learned during training. Concatenating all patterns forms the complete MPSE matrix

$$A = \text{concat}\left(A_{\text{scaled}}^{(1)}, A_{\text{scaled}}^{(2)}, \ldots, A_{\text{scaled}}^{(G)}\right) \in \mathbb{R}^{d_{\text{inner}} \times d_s}. \tag{4}$$

This matrix provides multiple parallel state evolution patterns at each time step, endowing the model with multipattern modeling capability. Furthermore, the fused multipattern matrix is split into two streams in the network: one path directly influences the state update equation, affecting the hidden state h_t, allowing the model to capture multipattern dynamics at the state level; the other path is input to a linear transformation layer to modulate the final output or channel responses, enabling adaptive weighting at the output layer.

3) Projection and Normalization of Decay Matrix: To effectively apply the MPSE matrix to feature representations, we perform projection and normalization

$$A_{\text{proj}} = W_A \bar{A}_{\perp}, \quad A_{\text{norm}} = \sigma\left(A_{\text{proj}}\right) \tag{5}$$

where $\bar{A}_{\perp} = (1/d_s)\sum_{j=1}^{d_s} A_{(:,j)} \in \mathbb{R}^{d_{\text{inner}}}$ is the mean along the state dimension, $W_A \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{model}}}$ is the learnable projection matrix mapping the pattern matrix to the feature space, and $\sigma(\cdot)$ is the Sigmoid function, ensuring the values are in the range $[0, 1]$. The normalized A_{norm} serves as the MPSE weight vector.

4) Multipattern Residual Fusion: Finally, the normalized state evolution weights are fused with the input feature X

$$X' = X \odot A_{\text{norm}} + X \tag{6}$$

where $X \in \mathbb{R}^{B \times T \times d_{\text{model}}}$ denotes the input features, B is the batch size, T is the sequence length, and $\odot$ represents element-wise multiplication. This residual connection preserves the original features while introducing multipattern modulation, allowing the model to simultaneously capture fast-changing and slow-changing state evolution patterns, thus enhancing dynamic modeling capability.

B. Dual-Path Spectral–Spatial

To fully exploit both spatial and spectral features of hyperspectral data, we design a DP-SS structure, as shown in Fig. 10. The input tensor is $X \in \mathbb{R}^{B \times C \times H \times W}$, where B denotes batch size, C is the number of spectral channels, and H and W are the height and width of the input image. The tensor is simultaneously fed into the spectral and spatial paths for parallel modeling. The spectral path focuses on "spectral changes along spatial pixels," while the spatial path focuses on "spatial changes along spectral channels."

1) Spectral Path: The spectral path aims to capture variations of each spectral band across spatial pixels. First, the input patch is unfolded along the spatial dimension to form a sequence

$$X_{\text{spe_in}} = \text{reshape}\,(X) \in \mathbb{R}^{B \times C \times (H \cdot W)} \tag{7}$$

where $X_{\text{spe_in}}$ is the input sequence to the spectral path, with sequence length $H \cdot W$ and feature dimension C.

A linear mapping is applied for learnable feature transformation

$$X_{\text{spe1}} = \text{Linear}\left(X_{\text{spe_in}}\right) \in \mathbb{R}^{B \times C \times (H \cdot W)} \tag{8}$$

followed by sequence modeling using Mamba-MPSE

$$X_{\text{spe2}} = \text{Mamba-MPSE}\left(X_{\text{spe1}}\right) \in \mathbb{R}^{B \times C \times C} \tag{9}$$

where the sequence length is adjusted from $H \cdot W$ to C while preserving complete spectral information. Finally, the sequence is mapped back and reshaped to the original spatial dimensions

$$X_{\text{spe_out}} = \text{reshape}\left(\text{Linear}\left(X_{\text{spe2}}\right)\right) \in \mathbb{R}^{B \times C \times H \times W}. \tag{10}$$

2) Spatial Path: The spatial path captures variations of each spatial pixel along spectral channels. The input patch is unfolded along the spectral dimension

$$X_{\text{spa_in}} = \text{reshape}\,(X) \in \mathbb{R}^{B \times (H \cdot W) \times C}. \tag{11}$$

A linear transformation is applied

$$X_{\text{spa1}} = \text{Linear}\left(X_{\text{spa_in}}\right) \in \mathbb{R}^{B \times (H \cdot W) \times C} \tag{12}$$

followed by Mamba-MPSE modeling

$$X_{\text{spa2}} = \text{Mamba-MPSE}\left(X_{\text{spa1}}\right) \in \mathbb{R}^{B \times (H \cdot W) \times C} \tag{13}$$

and finally reshaped back to the original dimensions

$$X_{\text{spa_out}} = \text{reshape}\left(X_{\text{spa2}}\right) \in \mathbb{R}^{B \times C \times H \times W}. \tag{14}$$

3) Sample-Level Differential Modulation: In hyperspectral image classification, each training batch often contains multiple classes. Directly applying a unified state evolution risks coupling different classes, weakening intraclass heterogeneity modeling. To address this, we introduce SDM based on MPSE to provide each sample with independent dynamic evolution strength, allowing adaptive integration of historical information.

Given input tensor $X \in \mathbb{R}^{B \times C \times H \times W}$, intermediate features are obtained via DP-SS linear mapping $F()$

$$f_i = F\left(X_i\right) \in \mathbb{R}^d \tag{15}$$

where d is the sequence length of the corresponding path. A conditional vector c_i is embedded from the label and passed through a conditional MLP $g()$ to compute dynamic evolution weights. The resulting coefficients α_i in SDM act as sample-level modulation factors, adjusting the effective strength of the MPSE state evolution matrix for each sample; in contrast, the learnable scaling factors α_i in MPSE define pattern-level decay rates representing intrinsic evolution dynamics shared across

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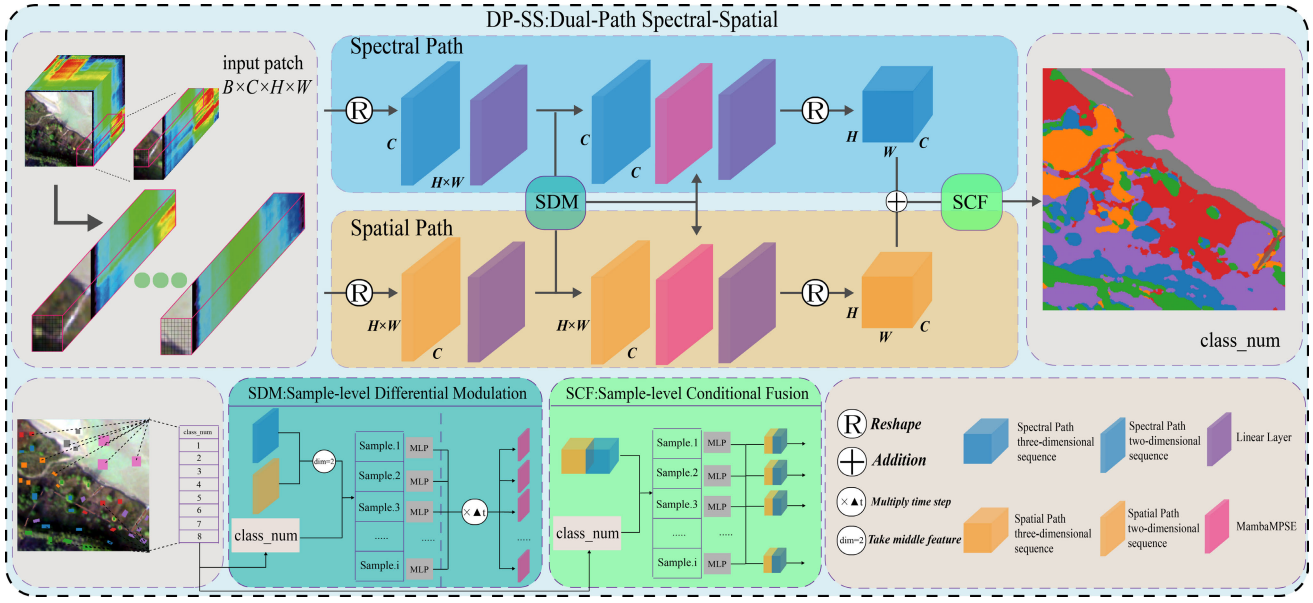


Fig. 10. Illustration of the DP-SS structure. The input hyperspectral block of size $B \times C \times H \times W$ is reshaped and fed into the spectral and spatial paths. Each path processes the features through linear layers, Mamba-MPSE, and SDM modules. The two path outputs are summed and further fused via the SCF module to produce the final class predictions. The legend below illustrates the operations and module representations.

all samples. Both types of α_i are multiplicative coefficients that modulate state evolution

$$\alpha_i = g\left(f_i, c_i\right) \in \mathbb{R}^k \tag{16}$$

where k is the internal MPSE weight dimension. These coefficients modulate the effective strength of the MPSE state evolution matrix

$$A_i = A \odot \sigma\left(\alpha_i\right) \tag{17}$$

where A is the original state evolution matrix, $\odot$ denotes element-wise multiplication, and σ is the sigmoid function. The updated hidden state for sample i becomes

$$h_t^{(i)} = \mathrm{MPSE}\left(h_{t-1}^{(i)}, x_t^{(i)}; \alpha_i\right) \tag{18}$$

allowing independent, adaptive evolution trajectories for each sample.

4) Sample-Level Conditional Fusion: After DP-SS feature extraction, we obtain spectral output $X_{\mathrm{spe_out}}$ and spatial output $X_{\mathrm{spa_out}}$. To adaptively combine features for each sample, we compute fusion weights. Global average pooling (GAP) extracts semantic descriptors

$$V_{\mathrm{spe}} = \mathrm{GAP}\left(X_{\mathrm{spe_out}}\right), \quad V_{\mathrm{spa}} = \mathrm{GAP}\left(X_{\mathrm{spa_out}}\right) \tag{19}$$

and concatenation yields

$$V_{\mathrm{cat}} = \left[V_{\mathrm{spe}}; V_{\mathrm{spa}}\right] \in \mathbb{R}^{B \times 2C}. \tag{20}$$

The sample-level fusion weight β_i is predicted using a conditional MLP

$$\beta_i = \sigma\left(g\left(V_{\mathrm{cat}}^{(i)}, c_i\right)\right) \in [0, 1]. \tag{21}$$

Fusion of spectral and spatial paths

$$X_{\mathrm{fused}}^{(i)} = \beta_i \cdot X_{\mathrm{spe_out}}^{(i)} + (1 - \beta_i) \cdot X_{\mathrm{spa_out}}^{(i)}. \tag{22}$$

A regularization term maintains balance

$$L_{\mathrm{fusion}} = \frac{1}{B} \sum_{i=1}^{B} (\beta_i - 0.5)^2. \tag{23}$$

Finally, the fused features are aggregated, normalized, and fed into the classifier to obtain predictions

$$X_{\mathrm{pool}} = \mathrm{GAP}\left(X_{\mathrm{fused}}\right) \tag{24}$$

$$X_{\mathrm{norm}} = \mathrm{LayerNorm}\left(\mathrm{flatten}\left(X_{\mathrm{pool}}, \mathrm{start_dim} = 1\right)\right) \tag{25}$$

$$Y = \mathrm{Linear}\left(X_{\mathrm{norm}}\right) \in \mathbb{R}^{B \times \mathrm{class_num}} \tag{26}$$

where Y represents the final class predictions for the batch.

IV. Experiments

A. Public Datasets Description

In addition to the self-collected UAV hyperspectral dataset of the Yellow River Wetlands, this study also employs a set of publicly available datasets highly relevant to wetland land cover classification tasks, following a "primary + auxiliary" data design framework consistent with the UAV dataset. The primary public satellite dataset, referred to as the Yellow River Estuary dataset, was acquired by the GF-5 satellite over the Yellow River Delta region. The image has a spatial size of 1185 × 1342 pixels with a spatial resolution of 30 m, containing 285 spectral bands and covering 20 land cover classes. The Yellow River Estuary dataset has a data volume of approximately 1.33 GB and is stored in floating-point format.

As the auxiliary public satellite dataset, the Botswana dataset complements the primary one to enhance the generalization capability of model evaluation. This dataset was collected by NASA's EO-1 satellite over the Okavango Delta in Botswana on May 31, 2001, with a spatial size of 1476 × 256 pixels, a spatial resolution of 30 m, containing 145 spectral bands, and covering 14 land cover classes.

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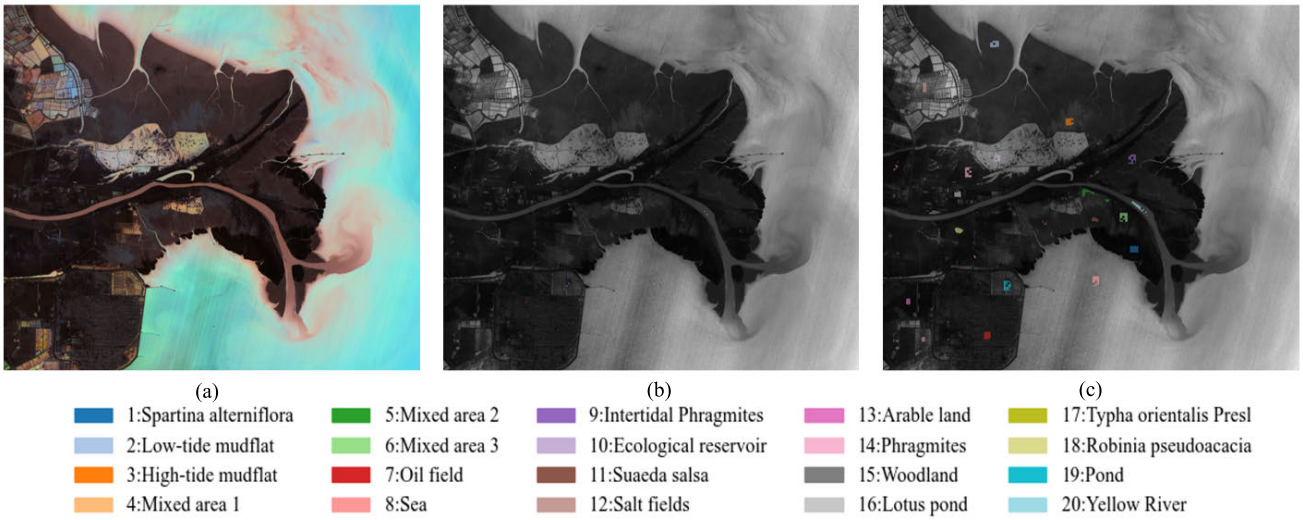


Fig. 11. False-color image of the Yellow River Estuary hyperspectral dataset and the corresponding sample distribution maps. (a) False-color image. (b) Distribution of training samples. (c) Distribution of testing samples. The bottom legend indicates the corresponding land cover types represented by specific label colors.

TABLE IV
SAMPLE NUMBERS OF TRAINING AND TESTING SETS ON THE YELLOW RIVER ESTUARY DATASET

Yellow River Estuary			
Class No	Class Name	Train	Test
1	Spartina alterniflora	56	1064
2	Low-tide mudflat	53	1007
3	High-tide mudflat	51	969
4	Mixed area 1	35	665
5	Mixed area 2	60	1140
6	Mixed area 3	58	1102
7	Oil field	48	912
8	Sea	62	1178
9	Intertidal Phragmites	58	1102
10	Ecological reservoir	46	874
11	Suaeda salsa	44	836
12	Salt fields	47	893
13	Arable land	51	969
14	Phragmites	59	1121
15	Woodland	36	684
16	Lotus pond	40	760
17	Typha orientalis Presl	38	722
18	Robinia pseudoacacia	42	798
19	Pond	54	1026
20	Yellow River	61	1159
Total		**999**	**18981**

TABLE V
SAMPLE NUMBERS OF TRAINING AND TESTING SETS ON THE BOTSWANA DATASET

Botswana			
Class No	Class Name	Train	Test
1	Water	13	257
2	Hippo grass	5	96
3	Floodplain grasses 1	12	239
4	Floodplain grasses 2	10	205
5	Reeds	13	256
6	Riparian	13	256
7	Firescar	12	247
8	Island interior	10	193
9	Acacia woodland	15	299
10	Acacia shrubland	12	236
11	Acacia grassland	15	290
12	Short mopane	9	172
13	Mixed mopane	13	255
14	Exposed soils	4	91
Total		**156**	**3092**

The Botswana dataset has a data volume of approximately 75.2 MB and is also stored in floating-point format.

For dataset partitioning, the same unified random sampling strategy as that used for the self-constructed Yellow River Wetland UAV dataset was adopted, where 5% of the labeled samples in both datasets were used for training, and the remaining 95% were used for testing. Tables IV and V present the numbers of training and testing samples for each land cover class in the Yellow River Estuary dataset and the Botswana dataset, respectively, while Figs. 11 and 12 illustrate the false-color images and corresponding sample distribution maps of the two datasets.

B. Experimental Settings

1) Evaluation Metrics: In addition to evaluating classification accuracy for each individual class, three quantitative metrics were used to analyze and assess model performance: overall accuracy (OA), average accuracy (AA), and Kappa coefficient. Higher values of per-class accuracy, OA, AA, and Kappa indicate better classification performance of the model.

2) Implementation Details: All experiments were conducted using a GeForce RTX 4090 GPU and implemented in PyTorch.

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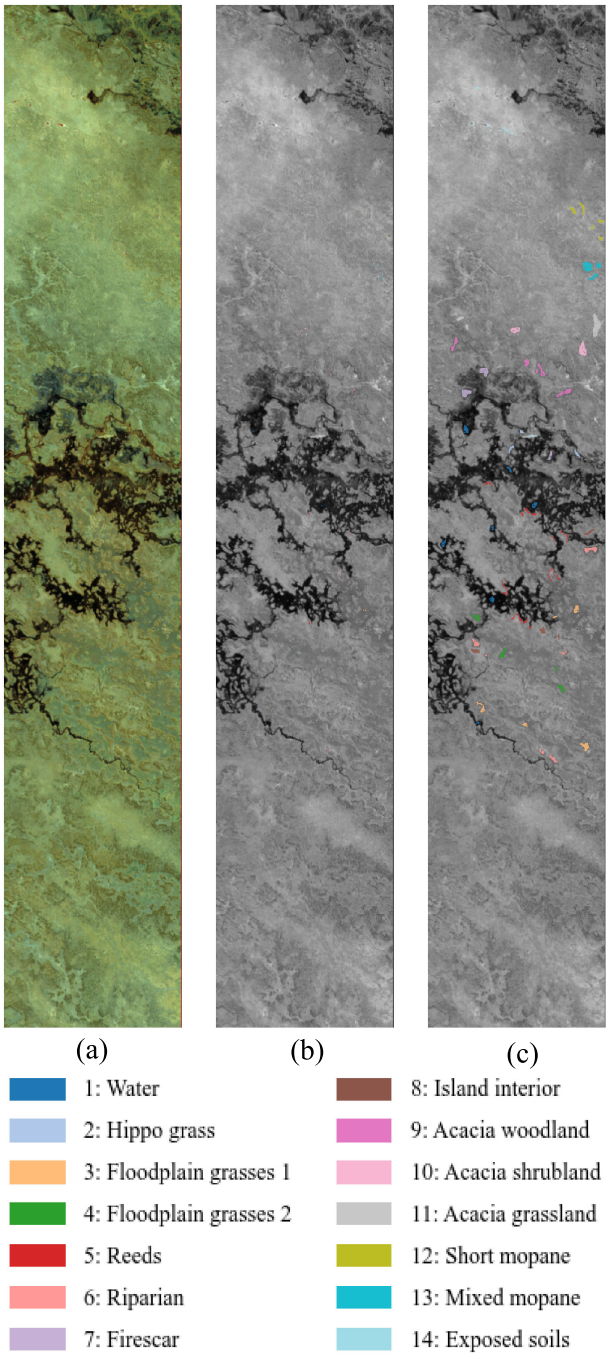


Fig. 12. False-color image of the Botswana hyperspectral dataset and the corresponding sample distribution maps. (a) False-color image. (b) Distribution of training samples. (c) Distribution of testing samples. The bottom legend indicates the corresponding land cover types represented by specific label colors.

To ensure the reliability and consistency of the results, the number of training epochs was set to 200. The batch size and patch size were fixed at 256 and 11, respectively. The model was trained using the Adam optimizer with an initial learning rate of 0.001 and a weight decay of 1×10^{-4} to prevent overfitting. Moreover, the core Mamba-MPSE module was configured with three pattern groups and a hidden dimension D of 150. To reduce the effect of randomness, each model was independently trained five times, and the average performance across these runs was reported as the final result. Specifically, the results are presented in the form of mean $\pm$ standard deviation. To address the limited number of training samples, data augmentation techniques were applied during training, including random rotation, horizontal and vertical flipping, and the addition of Gaussian noise.

3) Competitive Approaches: The study selected three representative HSI classification architectures for comprehensive comparison, including CNN-based methods (2D-CNN [34] and 3D-CNN [35]), transformer-based methods (MSTNet [36], LESSFormer [37], and MASSFormer [38]), Mamba-based methods (MambaHSI [17], 3DSS-Mamba [18], HyperMamba [19], and HSI-MFormer [20]), as well as the proposed method, Mamba-MPSE.

C. Experimental Comparison With Comparative Methods

The quantitative accuracies, including OA (%), AA (%), and Kappa (%), for the Yellow River Wetland (autumn) and Yellow River Estuary datasets are summarized in Tables VI and VII, with the best results highlighted in bold. The corresponding classification maps are shown in Figs. 13 and 14.

1) Yellow River Wetland (Autumn) Dataset Results Analysis: The classification results of different methods on the Yellow River Wetland (autumn) are presented in Fig. 13 and Table VI. Among CNN-based approaches, 2D-CNN and 3D-CNN are adopted as baseline methods. By relying on convolution operations to extract local spatial features, CNN-based models perform stably in high-accuracy classes such as Yellow River and Mudflat, both achieving nearly 100% accuracy. However, their performance degrades noticeably in medium- and low-accuracy classes, including Road (60.50% for 2D-CNN and 86.04% for 3D-CNN) and Cogon grass (69.98% and 75.65%, respectively). The overall OA of 2D-CNN and 3D-CNN are 84.82% and 85.27%, respectively, while the corresponding AA are 91.23% and 92.96%, with Kappa coefficients of 82.61% and 91.67%. These results indicate that conventional convolutions are effective for local texture extraction but insufficient for modeling global spectral sequences and cross-spatial pixel dependencies.

For transformer-based methods, including MSTNet, LESSFormer, and MASSFormer, self-attention mechanisms and multilayer encoding strategies are introduced to enhance long-range spectral–spatial dependency modeling. MSTNet adopts a multilevel spectral–spatial transformer encoder to fuse sequential features with positional encoding, achieving an OA of 95.68% and an AA of 95.97%. It maintains nearly 100% accuracy in high-accuracy classes while significantly improving medium- and low-accuracy classes, such as Cogon grass and road, to 91.52% and 88.68%, respectively. LESSFormer employs the HSI2Token module and locally enhanced transformer encoding to strengthen local feature representation while preserving global contextual information, resulting in more stable classification of Miscanthus (95.39%) and Cogon grass (89.56%). MASSFormer further incorporates a memory-enhanced module to optimize spectral–spatial information fusion, achieving balanced performance on road (89.68%) and Cogon grass (94.48%), with an overall OA of 96.46%

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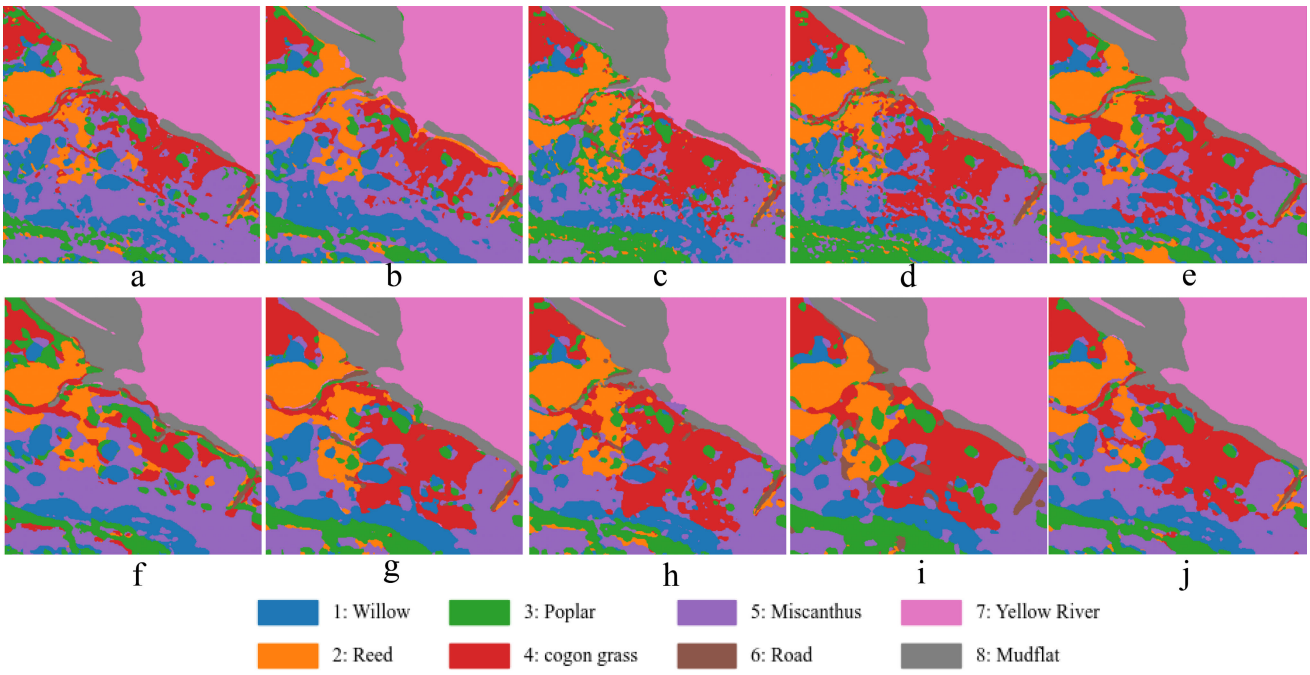


Fig. 13. Visualization of classification results on the Yellow River Wetland (autumn) dataset. (a) 2D-CNN result, (b) 3D-CNN result, (c) MSTNet result, (d) LESSFormer result, (e) MASSFormer result, (f) MambaHSI result, (g) 3DSS-Mamba result, (h) HyperMamba result, (i) HSI-MFormer result, and (j) Mamba-MPSE result. The bottom legend indicates the corresponding land cover types represented by specific label colors.

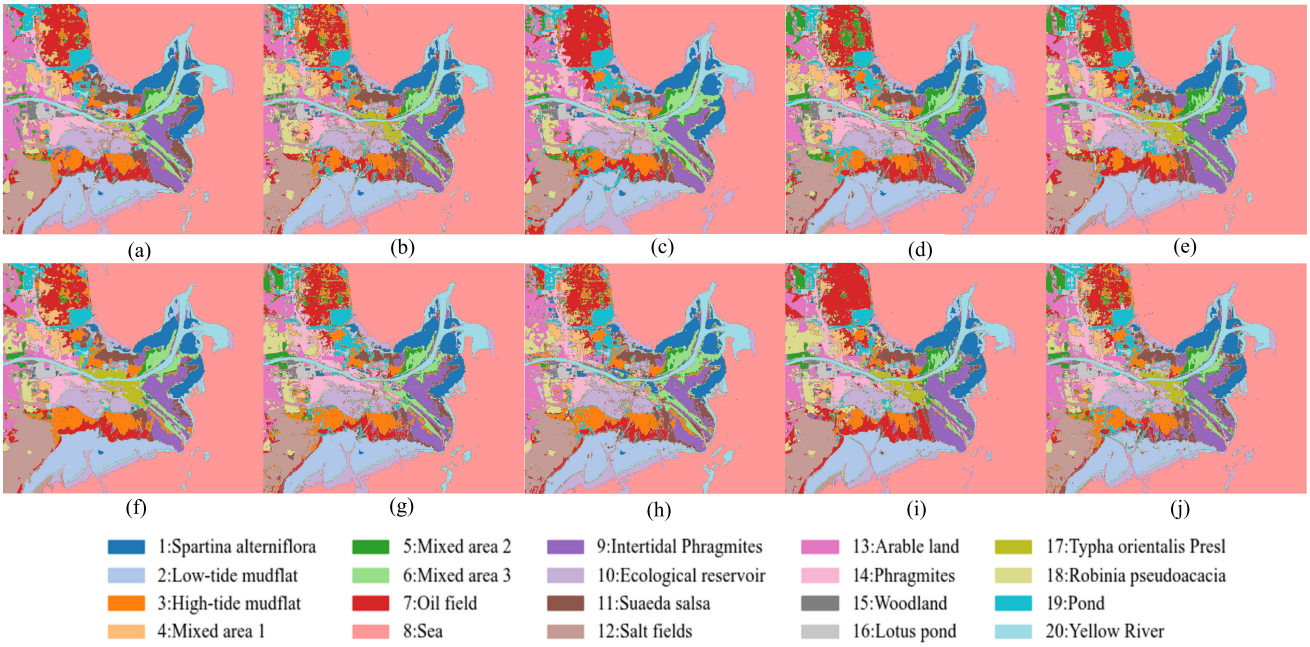


Fig. 14. Visualization of classification results on the Yellow River Estuary dataset. (a) 2D-CNN result, (b) 3D-CNN result, (c) MSTNet result, (d) LESSFormer result, (e) MASSFormer result, (f) MambaHSI result, (g) 3DSS-Mamba result, (h) HyperMamba result, (i) HSI-MFormer result, and (j) Mamba-MPSE result. The bottom legend indicates the corresponding land cover types represented by specific label colors.

and AA of 96.97%. This design effectively mitigates feature attenuation in deep transformer layers.

For Mamba-based methods, including MambaHSI, 3DSS-Mamba, and HyperMamba, SSMs are employed to enable global sequence modeling with linear computational complexity. MambaHSI integrates spectral and SpaMB blocks with fusion modules to realize pixel-level long-range interactions, substantially improving the classification accuracy of road (87.56%) and Cogon grass (93.58%). It achieves an overall OA of 96.47%, an AA of 96.72%, and a Kappa coefficient of 96.95%. 3DSS-Mamba introduces 3-D spectral–spatial selective scanning and multidirectional strategies, maintaining nearly 100% accuracy in high-accuracy classes while further stabilizing medium-accuracy classes such as

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TABLE VI

CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON YELLOW RIVER WETLAND (AUTUMN) DATASET. BOLD NUMBERS INDICATE THE BEST RESULTS

Class/Methods	CNN-based		Transformer-based			Mamba-based				
	2D-CNN [34]	3D-CNN [35]	MSTNet [36]	LESSFormer [37]	MASSFormer [38]	MambaHSI [17]	3DSS-Mamba [18]	HyperMamba [19]	HSI-MFormer [20]	Mamba-MPSE
Willow	85.87±2.25	95.25±1.02	96.58±0.27	97.98±0.87	98.25±0.57	98.33±0.76	97.93±0.95	97.37±1.02	98.72±0.81	**98.92±0.87**
Reed	94.22±1.13	97.87±0.93	98.94±0.98	99.29±0.12	98.88±0.86	99.36±0.07	98.76±1.11	99.12±0.41	**99.63±0.08**	99.58±0.14
Poplar	87.14±2.37	89.51±1.92	97.32±1.08	96.57±0.36	96.19±1.17	95.49±1.81	96.26±1.63	97.02±0.69	97.89±1.78	**98.79±1.03**
Cogon grass	69.98±7.85	75.65±3.25	91.52±1.06	89.56±1.79	94.48±1.34	93.58±1.91	95.54±1.75	94.68±1.76	96.12±1.21	**97.92±0.86**
Miscanthus	82.57±3.01	85.55±2.28	92.36±1.78	95.39±0.57	94.18±1.52	97.46±1.24	97.29±0.79	**98.69±1.07**	97.98±0.92	98.61±1.14
Road	60.50±9.74	86.04±3.98	88.68±1.76	90.56±1.99	89.68±2.36	87.56±2.71	90.74±1.13	89.75±2.38	90.65±1.98	**92.36±1.75**
Yellow River	99.54±0.19	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Mudflat	98.77±1.02	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA	84.82±2.64	91.23±2.36	95.68±1.29	96.17±1.37	96.46±1.55	96.47±1.82	97.06±1.09	97.08±0.86	97.62±0.91	**98.27±0.83**
AA	85.27±2.08	92.96±1.54	95.97±1.88	97.15±1.19	96.97±1.41	96.72±1.01	97.82±0.94	97.95±1.63	98.34±1.49	**98.84±1.11**
Kappa	82.61±2.96	91.67±1.97	96.26±1.97	96.64±1.08	96.81±1.39	96.95±1.25	97.42±1.32	97.57±1.05	98.04±1.09	**98.63±1.54**
Parameter Count (K)	25	60	248	272	255	82	105	94	156	102
FLOPs (G)	12.6	48.3	526.7	601.5	558.2	152.4	201.8	176.3	342.9	189.7
Inference Speed (ms/sample)	0.52	1.89	6.42	7.95	7.18	2.45	3.15	2.88	4.83	3.05

TABLE VII

CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON YELLOW RIVER ESTUARY DATASET. BOLD NUMBERS INDICATE THE BEST RESULTS

Class/Methods	CNN-based		Transformer-based			Mamba-based				
	2D-CNN [34]	3D-CNN [35]	MSTNet [36]	LESSFormer [37]	MASSFormer [38]	MambaHSI [17]	3DSS-Mamba [18]	HyperMamba [19]	HSI-MFormer [20]	Mamba-MPSE
Class 1	98.59±0.99	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 2	91.75±2.18	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.09±1.12	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 3	94.42±1.91	99.22±0.36	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 4	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 5	36.72±8.27	47.98±7.15	49.15±7.43	55.98±6.33	52.36±6.45	53.07±6.08	50.15±7.21	54.47±6.66	**57.07±6.42**	56.98±6.11
Class 6	79.84±3.96	86.05±2.73	96.95±1.21	95.27±1.01	95.12±1.37	97.74±1.16	96.33±1.42	**99.85±0.09**	95.12±1.21	99.25±0.54
Class 7	92.23±1.97	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.10±0.28	99.84±0.10	98.34±0.96	97.81±1.18	99.97±0.01
Class 8	92.55±1.88	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 9	97.89±0.96	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 10	94.52±2.08	**100.00±0.00**	99.98±0.01	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	97.87±1.28	**100.00±0.00**	**100.00±0.00**
Class 11	68.56±4.99	75.57±4.08	84.87±2.39	87.65±2.64	89.02±1.24	90.85±1.82	91.80±1.43	82.79±2.87	89.02±1.49	**97.80±1.01**
Class 12	55.69±6.13	62.30±4.21	68.89±4.97	63.80±4.38	69.03±3.11	66.58±4.24	61.50±4.07	63.80±4.82	67.19±5.92	**74.07±3.23**
Class 13	89.81±2.31	98.97±0.55	98.05±0.95	99.41±0.22	98.62±0.71	**100.00±0.00**	**100.00±0.00**	99.41±0.11	98.57±0.81	**100.00±0.00**
Class 14	97.43±1.07	**100.00±0.00**	**100.00±0.00**	98.59±0.77	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 15	77.45±3.85	86.25±2.13	86.66±2.85	88.09±1.99	88.59±2.08	94.81±1.71	92.03±1.96	92.67±1.74	98.80±0.49	**100.00±0.00**
Class 16	74.56±3.03	87.64±2.04	87.32±2.01	88.63±2.95	89.98±1.86	**91.95±1.14**	91.87±1.94	78.09±3.38	84.60±3.74	90.45±1.77
Class 17	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 18	83.69±2.85	85.77±2.74	84.88±3.21	88.87±2.96	90.84±2.02	89.67±2.47	88.87±2.87	92.79±1.59	92.05±1.66	**98.50±0.91**
Class 19	96.36±1.48	97.90±1.57	99.59±0.16	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Class 20	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA	84.52±2.61	92.35±1.44	93.87±1.65	94.65±1.02	95.28±1.75	95.77±1.65	95.92±1.08	94.77±1.13	95.34±1.55	**97.09±1.27**
AA	84.79±2.71	90.13±1.98	94.11±1.48	94.92±1.67	95.54±1.51	96.01±1.11	96.18±1.04	95.11±1.24	95.61±1.39	**97.38±1.41**
Kappa	83.98±2.93	91.62±2.08	93.41±1.84	94.11±1.99	94.73±11.82	95.29±1.42	95.47±1.39	94.29±1.24	94.88±1.45	**96.55±1.22**
Parameter Count (K)	25	60	248	272	255	82	105	94	156	102
FLOPs (G)	268.1	1027.1	11211.8	12802.9	11883.5	3244.3	4294.3	3751.7	7297.9	4036.8
Inference Speed (ms/sample)	11.07	40.21	136.6	169.1	152.8	52.1	67.1	61.3	102.8	64.9

Miscanthus (97.29%) and Cogon grass (95.54%). HyperMamba adopts spatial neighborhood-adaptive scanning and spectral adaptive enhanced scanning, leading to additional improvements in medium-accuracy classes, including Road (89.75%), Cogon grass (94.68%), and Willow (97.37%), while preserving optimal performance in high-accuracy classes. These results highlight the strong capability of Mamba-based models in global sequence modeling and spectral–spatial feature integration.

Overall, by integrating the complementary strengths of CNN, transformer, and Mamba architectures, the proposed Mamba-MPSE achieves efficient DP-SS modeling via a multiscale temporal evolution mechanism. Experimental results on the Yellow River Wetland (autumn) dataset demonstrate that Mamba-MPSE attains an OA of 98.27%, an AA of 98.84%, and a Kappa coefficient of 98.63%, with stable performance on medium- and low-accuracy classes (e.g., road at 92.36%, Cogon grass at 97.92%) and 100% accuracy on high-accuracy classes, effectively unifying local feature extraction, long-range contextual modeling, and global sequence representation. Complexity analysis shows that 2D-/3D-CNN serve as lightweight baselines with limited modeling capability, transformer-based models suffer from high computational overhead and slow inference, while traditional Mamba-based models balance efficiency and performance. Mamba-MPSE, with only 102k parameters, 189.7-G FLOPs, and an inference speed of 3.05 ms/sample, achieves optimal accuracy with moderate complexity, highlighting the dual advantages of high precision and efficiency.

2) Yellow River Estuary Dataset Results Analysis: The classification results of different methods on the Yellow River Estuary dataset are presented in Fig. 14 and Table VII. Among CNN-based approaches, 2D-CNN and 3D-CNN are adopted as baseline methods. By relying on convolution operations for local feature extraction, CNN-based models perform stably in high-accuracy classes such as Spartina alterniflora, High-tide mudflat, and Mixed area 1, achieving 98.59%–100% for 2D-CNN and 99.22%–100% for 3D-CNN. However, their performance decreases noticeably in medium- and low-accuracy classes, including mixed area 2 (36.72% for 2D-CNN and 47.98% for 3D-CNN), salt fields (55.69% and 62.30%), and woodland (77.45% and 86.25%). The overall OA of 2D-CNN and 3D-CNN are 84.52% and 84.79%, respectively, while the corresponding AA are 92.35% and 90.13%, with Kappa coefficients of 83.98% and 91.62%. These results indicate that conventional convolutions are effective for local texture extraction but insufficient for modeling global spectral sequences and cross-spatial pixel dependencies.

For transformer-based methods, including MSTNet, LESSFormer, and MASSFormer, these transformer-based methods introduce self-attention and multilayer encoding to enhance long-range spectral–spatial dependency modeling. MSTNet

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employs multilevel spectral–spatial encoding, achieving an OA of 93.87% and an AA of 94.11%, with near-perfect accuracy in high-accuracy classes and notable improvements in medium- and low-accuracy classes such as mixed area 2 (49.15%) and salt fields (68.89%). LESSFormer further stabilizes mixed area 2 (55.98%) and salt fields (63.80%) while maintaining 100% accuracy in classes such as sea, oil field, and intertidal phragmites. MASSFormer incorporates a memory-enhanced module for spectral–spatial fusion, achieving more balanced performance (mixed area 2 52.36%, salt fields 69.03%, and woodland 88.59%) with an OA of 95.28% and an AA of 95.54%. These results demonstrate that transformer-based approaches effectively alleviate deep feature attenuation while capturing long-range dependencies.

For Mamba-based methods, including MambaHSI, 3DSS-Mamba, and HyperMamba, these SSMs enable global sequence modeling with linear computational complexity. MambaHSI combines spatial and SpeMB blocks with fusion modules, boosting mixed area 2 (53.07%), salt fields (66.58%), and woodland (94.81%) and achieving OA 95.77%, AA 96.01%, and Kappa 95.29%. 3DSS-Mamba employs 3-D spectral–spatial selective scanning and multidirectional strategies, preserving near-100% accuracy in high-accuracy classes and enhancing medium-accuracy classes such as mixed area 3 (96.33%) and woodland (92.03%). HyperMamba applies spatial neighborhood and spectral adaptive scanning, balancing medium- and low-accuracy classes (mixed area 2 54.47%, salt fields 63.80%, and woodland 92.67%) while maintaining optimal performance in high-accuracy classes (Spartina alterniflora, Sea, Phragmites, and Yellow River at 100%). These results demonstrate Mamba-based models' strong capability in global sequence modeling and spectral–spatial integration.

Overall, by integrating the complementary strengths of CNN, transformer, and Mamba architectures, the proposed Mamba-MPSE achieves efficient DP-SS modeling under a multiscale temporal evolution mechanism. Experimental results show that Mamba-MPSE attains an OA of 97.09%, an AA of 97.38%, and a Kappa coefficient of 96.55%. Medium- and low-accuracy classes (mixed area 2 56.98%, salt fields 74.07%, and woodland 100%) remain stable and balanced, while high-accuracy classes (Spartina alterniflora, Sea, Phragmites, and Yellow River) consistently reach 100%. Although computational cost increases proportionally with data scale, Mamba-MPSE maintains a favorable efficiency–accuracy tradeoff compared with transformer-based models whose complexity grows substantially under large-scale hyperspectral inputs. These results demonstrate its effectiveness in jointly modeling local features, long-range dependencies, and global sequence representations for hyperspectral image classification.

3) Summary: The proposed Mamba-MPSE method achieves excellent overall performance across both datasets. High-accuracy classes, such as Yellow River and Mudflat in the Yellow River Wetland (autumn) dataset, and Spartina alterniflora and High-tide mudflat in the Yellow River Estuary dataset, consistently reach near-100% accuracy due to relatively homogeneous spectral and spatial characteristics.

Medium- and low-accuracy classes, including road, Cogon grass, mixed area 2, and salt fields, show lower performance mainly because of intraclass heterogeneity. For example, road areas may contain mixtures of soil, grass, and water surfaces; Cogon grass and Miscanthus vary in growth conditions and soil background; and mixed areas and salt fields combine multiple land cover types, producing complex spectral–spatial patterns.

Mamba-MPSE effectively mitigates these challenges. The MPSE multipattern evolution captures diverse intraclass spectral–spatial variations, the DP-SS dual-path structure decouples spectral and spatial modeling, and the SDM/SCF modules balance contributions while avoiding interclass interference. Consequently, the method improves classification stability and accuracy for heterogeneous classes while maintaining near-perfect performance for homogeneous ones.

D. Auxiliary Dataset Experiments With Comparative Methods

The quantitative classification accuracies, including OA (%), AA (%), and Kappa (%), for the Yellow River Wetland (winter) and Botswana datasets are summarized in Tables VIII and IX, respectively, with the best results highlighted in bold.

1) Yellow River Wetland (Winter) Dataset Results Analysis: The classification results on the Yellow River Wetland (winter) dataset are shown in Table VIII. By using hyperspectral observations from a different season while keeping the scene unchanged, this experiment evaluates the generalization of the proposed model and implicitly considers seasonal hyperspectral variations. Overall, the performance ranking of methods in winter closely matches the autumn results. Mamba-MPSE achieves the best performance, with an OA of 98.47%, an AA of 97.86%, and a Kappa of 97.23%, demonstrating robust cross-seasonal capability.

Although overall differences are small, some class-level variations occur due to seasonal changes. Tree-related classes like Willow and Poplar show slightly reduced accuracy because of leaf senescence, while herbaceous species with distinct panicle structures, such as Miscanthus and Reed, show improved performance as their spectral–spatial features become more concentrated. Reduced vegetation coverage also lessens occlusion for the Road class, improving accuracy. Spectrally stable classes like Yellow River and Mudflat consistently maintain near-perfect classification.

These results indicate that winter observations do not significantly affect overall performance; yet, they provide complementary evidence of the adaptability of Mamba-MPSE to seasonal variations. Moreover, the method maintains moderate computational complexity while achieving high classification accuracy without incurring excessive FLOPs or inference time.

2) Botswana Dataset Results Analysis: The classification results on the Botswana dataset are presented in Table IX. To assess the generalization of the proposed method across geographic regions and data distributions, an auxiliary experiment was conducted using this publicly available hyperspectral dataset. Despite notable differences from the primary Yellow River Estuary dataset in scene characteristics and data scale, Mamba-MPSE achieves

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TABLE VIII

CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON YELLOW RIVER WETLAND (WINTER) DATASET. BOLD NUMBERS INDICATE THE BEST RESULTS

Class/Methods	CNN-based		Transformer-based			Mamba-based				
	2D-CNN [34]	3D-CNN [35]	MSTNet [36]	LESSFormer [37]	MASSFormer [38]	MambaHSI [17]	3DSS-Mamba [18]	HyperMamba [19]	HSI-MFormer [20]	Mamba-MPSE
Willow	83.92±2.31	93.18±1.15	94.86±0.39	95.12±0.93	95.48±0.71	95.67±0.88	94.21±1.02	95.74±1.08	95.89±0.92	**96.45±1.02**
Reed	93.65±1.24	97.42±0.98	98.51±0.86	99.06±0.21	98.64±0.92	99.12±0.15	98.93±0.87	98.26±0.48	**99.82±0.04**	99.15±0.17
Poplar	84.96±2.61	87.42±2.03	94.68±1.19	94.92±0.44	94.61±1.28	93.88±1.93	94.73±1.71	95.61±0.82	94.32±1.84	**95.98±1.21**
Cogon grass	70.86±7.42	76.94±3.11	92.38±1.12	90.74±1.65	95.36±1.21	94.21±1.64	96.02±1.43	95.27±1.59	96.74±1.09	**97.10±1.05**
Miscanthus	83.46±2.88	86.94±2.14	93.58±1.62	96.42±0.61	95.37±1.29	98.12±1.03	97.88±0.72	98.41±0.76	98.24±0.86	**99.32±0.21**
Road	64.21±9.12	88.36±3.61	90.21±1.54	92.03±1.73	91.12±2.04	89.74±2.28	92.36±1.04	91.48±2.12	92.27±1.74	**94.88±1.24**
Yellow River	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Mudflat	99.17±0.22	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA	85.78±2.41	92.03±2.12	96.21±1.34	96.68±1.22	96.94±1.31	96.98±1.47	97.42±1.08	97.56±0.96	97.89±0.93	**98.47±0.81**
AA	85.03±2.65	91.28±1.71	95.53±1.24	96.04±1.45	96.32±1.42	96.34±1.61	96.77±1.09	96.85±1.23	97.16±1.33	**97.86±1.32**
Kappa	83.96±2.73	90.81±2.26	95.02±1.49	95.61±1.36	95.87±1.44	95.93±1.58	96.28±1.21	96.41±1.09	96.74±1.06	**97.23±0.94**
Parameter Count (K)	25	60	248	272	255	82	105	94	156	102
FLOPs (G)	12.5	47.7	520.1	594.6	551.3	150.6	199.5	174.1	338.7	187.5
Inference Speed (ms/sample)	0.51	1.87	6.34	7.86	7.11	2.42	3.11	2.84	4.77	3.01

TABLE IX

CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON BOTSWANA DATASET. BOLD NUMBERS INDICATE THE BEST RESULTS

Class/Methods	CNN-based		Transformer-based			Mamba-based				
	2D-CNN [34]	3D-CNN [35]	MSTNet [36]	LESSFormer [37]	MASSFormer [38]	MambaHSI [17]	3DSS-Mamba [18]	HyperMamba [19]	HSI-MFormer [20]	Mamba-MPSE
Class 1	82.14±2.42	84.36±2.09	85.72±2.68	87.91±2.21	87.48±2.34	86.63±2.05	87.94±1.87	89.71±1.82	90.36±1.67	**91.82±1.44**
Class 2	66.83±5.51	70.24±4.76	71.62±3.84	76.95±2.29	74.38±3.11	75.86±2.74	76.91±2.46	78.24±2.38	77.62±2.29	**79.14±2.05**
Class 3	71.34±4.98	75.92±4.06	79.86±3.17	82.41±2.62	81.73±2.78	80.94±2.53	82.16±2.28	81.87±2.19	82.03±2.07	**83.02±1.92**
Class 4	68.57±5.31	72.86±4.44	76.94±3.41	78.26±3.02	78.81±2.97	79.48±2.58	80.61±2.31	80.44±2.24	81.03±2.12	**81.94±1.87**
Class 5	73.68±4.63	77.52±3.88	82.74±2.69	83.96±2.41	83.54±2.53	84.62±2.13	85.37±1.98	85.14±1.91	**86.83±1.74**	86.41±1.66
Class 6	71.92±4.74	75.63±4.02	85.92±1.08	84.41±1.53	83.17±2.61	83.26±2.19	84.43±1.97	84.21±1.91	84.96±1.78	**86.12±1.53**
Class 7	80.86±2.03	85.14±2.47	86.37±2.11	88.62±1.32	86.18±2.21	88.91±1.93	89.64±1.81	88.21±1.58	88.63±1.52	**91.94±1.47**
Class 8	73.03±4.18	74.92±3.36	79.31±2.27	80.84±2.91	82.53±1.02	83.72±1.49	**85.91±1.23**	83.34±1.16	83.67±1.04	85.28±1.08
Class 9	75.94±4.42	79.81±3.79	85.16±2.04	84.62±2.51	84.27±2.59	87.61±1.11	86.48±1.94	87.21±1.88	86.83±1.74	**88.47±1.51**
Class 10	73.16±4.69	76.54±3.98	81.23±2.92	82.41±2.61	82.96±2.47	83.74±2.18	84.86±1.94	82.61±1.87	83.12±1.76	**86.73±1.53**
Class 11	76.92±4.21	80.37±3.61	86.63±1.56	83.92±2.29	87.31±1.17	86.94±1.96	87.82±1.78	**88.91±1.71**	88.64±1.55	88.13±1.47
Class 12	69.74±4.97	73.61±4.13	78.42±3.12	80.81±2.79	81.43±2.87	84.52±2.34	85.64±2.11	83.32±2.04	85.06±1.92	**86.74±1.68**
Class 13	71.84±4.76	75.36±3.99	80.13±2.97	81.62±2.63	84.27±1.74	83.36±1.29	85.08±1.05	84.11±1.98	**85.74±1.87**	85.26±1.62
Class 14	64.91±6.02	68.87±5.21	73.16±3.91	74.92±3.46	74.31±3.58	75.86±3.02	76.94±2.76	76.58±2.69	77.41±2.53	**79.12±2.24**
OA	73.89±2.74	77.48±2.43	82.12±2.01	83.10±1.78	83.31±1.86	84.14±1.61	85.11±1.47	84.72±1.41	85.29±1.33	**86.52±1.14**
AA	72.92±2.93	76.51±2.61	80.94±2.13	82.26±1.89	82.31±1.97	83.25±1.69	84.27±1.54	83.85±1.49	84.42±1.41	**85.72±1.21**
Kappa	71.71±3.11	75.61±2.78	80.63±2.21	81.69±1.97	81.91±2.04	82.81±1.82	83.85±1.66	83.45±1.61	84.07±1.52	**85.39±1.31**
Parameter Count (K)	25	60	248	272	255	82	105	94	156	102
FLOPs (G)	15.2	58.1	633.5	723.0	671.2	183.3	242.8	212.0	412.1	228.3
Inference Speed (ms/sample)	0.63	2.27	7.72	9.57	8.64	2.95	3.79	3.46	5.81	3.67

TABLE X

ABLATION EXPERIMENT RESULTS ON DIFFERENT WETLAND DATASETS

Original Mamba	MPSE	DP-SS		OA (%)	
		Spe Path	Spa Path	YRW	YRE
✓	×	×	×	87.58	88.74
×	✓	×	×	90.81	91.52
✓	×	✓	×	93.37	94.03
×	✓	✓	×	95.79	95.16
×	✓	×	✓	96.04	95.88
✓	×	✓	✓	96.68	95.34
×	✓	✓	✓	98.27	97.09

the best performance, with OA, AA, and Kappa of 86.52%, 85.72%, and 85.39%, demonstrating strong robustness and generalization.

OA is lower than that on the primary dataset, mainly due to the limited labeled samples (only 5% used for training) and the small-sample scenario. In contrast, the Yellow River dataset has more samples, richer spectral information, and more complex land cover distributions, supporting stronger feature learning and higher accuracy.

Under these challenging conditions, all methods show some performance decline; yet, Mamba-MPSE maintains stable and balanced classification across classes, outperforming CNN, transformer, and traditional Mamba methods in both single-class accuracy and overall metrics. This indicates strong adaptability to limited training data and reduced spectral information, validating its effectiveness in small-sample hyperspectral classification.

Overall, the Botswana experiment serves as auxiliary validation, reinforcing the generalization of Mamba-MPSE across data scales, sensor types, and geographic environments. Similarly, the method maintains moderate computational complexity while achieving high classification accuracy without incurring excessive FLOPs or inference time.

E. Ablation Study

1) Experiments: To comprehensively evaluate the contribution of each module to classification performance, we strictly followed the implementation details described above to ensure fairness. Seven groups of ablation experiments were designed, using OA as the core evaluation metric. In addition to the Yellow River Wetland (autumn) dataset, the same ablation settings were further evaluated on the Yellow River Estuary dataset to assess cross-regional robustness, where YRW and YRE denote the Yellow River Wetland (autumn) and Yellow River Estuary datasets, respectively. For brevity, the spectral path and spatial path in DP-SS are abbreviated as Spe Path and Spa Path. The seven groups are configured as follows. Group 1 uses only the original Mamba. Group 2 uses only MPSE. Group 3 combines the original Mamba with the Spe Path in DP-SS, without MPSE or the Spa Path. Group 4 combines MPSE with the Spe Path, without the original Mamba or Spa Path. Group 5 combines MPSE with the Spa Path, without the original Mamba or Spe Path. Group 6 combines the original Mamba with full DP-SS, without MPSE. Group 7 combines MPSE with full DP-SS, without the original Mamba.

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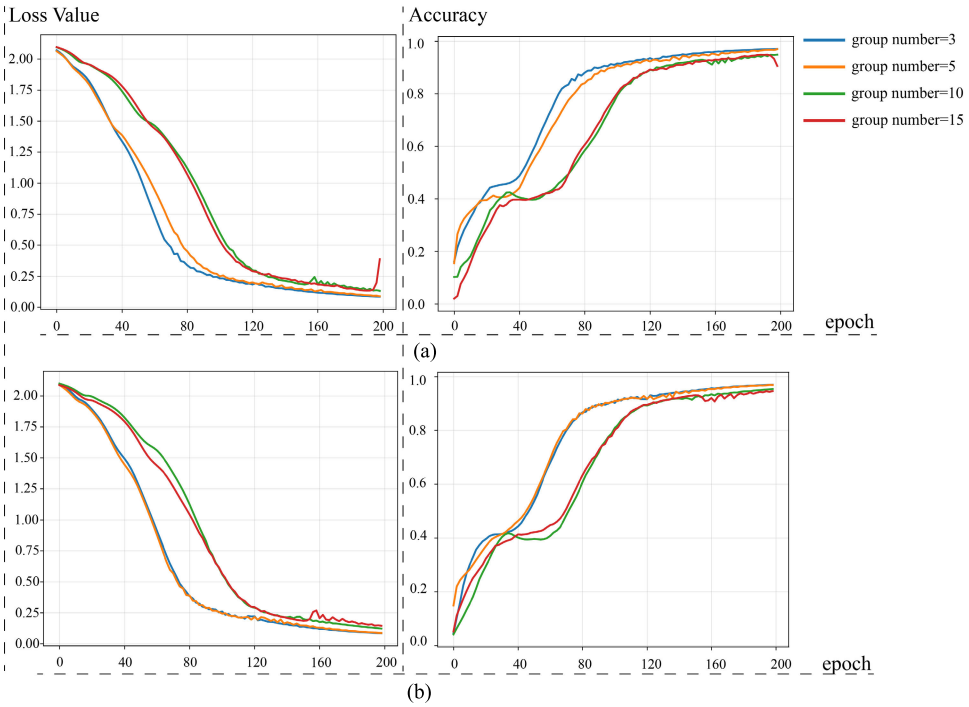


Fig. 15. This figure illustrates the loss and accuracy curves over 200 training epochs for group numbers of 3, 5, 10, and 15 on the Yellow River Wetland (autumn) dataset and the Yellow River Estuary dataset. The left part shows the loss versus training epoch curves for different group numbers, while the right part shows the corresponding accuracy versus training epoch curves. The curve colors follow the legend (blue: group number = 3, orange: group number = 5, green: group number = 10, red: group number = 15). (a) Yellow River Wetland (autumn) dataset. (b) Yellow River Estuary dataset.

The corresponding OAs on YRW and YRE are reported in Table X.

2) Discussion: Comparing Groups 1 and 2 shows that introducing MPSE consistently outperforms the original Mamba on both YRW and YRE, highlighting its superior ability to capture complex temporal dynamics of land cover features. Results further indicate that the Spe Path and Spa Path in DP-SS provide complementary gains, with the Spa Path exhibiting more stable improvements across datasets. Finally, the synergy between MPSE and DP-SS (Group 7) achieves the largest performance boost on both datasets, demonstrating that the proposed module combination not only enhances temporal modeling but also ensures strong cross-regional robustness and environmental adaptability.

F. Hyperparameter Experiment

1) Experiments: To evaluate the impact of different group numbers in the MPSE mechanism on classification performance, this experiment fixed all other parameters and only adjusted the group number G (set to 3, 5, 10, 15, with G = 3 as the baseline). Training was performed for 200 epochs on both the Yellow River Wetland (autumn) dataset and the Yellow River Estuary dataset, and the loss and accuracy curves are shown in Fig. 15.

For the Yellow River Wetland (autumn) dataset in subplot a, the loss curves show that in the initial stage (first 40 epochs), G = 3 (blue) decreased fastest while G = 15 (red) decreased slowest. In the middle stage (40–160 epochs), G = 3 continued a steady decline approaching zero, whereas the other groups decreased more slowly and G = 15 exhibited slight fluctuations. In the later stage (after 160 epochs), G = 3 stabilized at a very low level, while the others stabilized at significantly higher loss values. For the accuracy curves, in the initial stage (first 20 epochs), G = 3 rose fastest, quickly surpassing 0.4, whereas G = 15 increased most slowly. In the middle stage (20–100 epochs), G = 3 continued rising rapidly toward 1.0, G = 5 increased moderately, and G = 10 and G = 15 rose slowly. In the later stage (after 100 epochs), G = 3 stabilized near 1.0 with minimal fluctuation, G = 5 also stabilized but at a lower accuracy, and G = 10 and G = 15 continued slow growth remaining below G = 3.

For the Yellow River Estuary dataset in subplot b, the loss curves show that in the initial stage (first 40 epochs), G = 3 and G = 5 decreased at similar rates, both faster than G = 10 and G = 15. In the middle stage (40–160 epochs), G = 3 and G = 5 continued to decline steadily with minimal differences in loss values, while G = 10 and G = 15 decreased more slowly. In the later stage (after 160 epochs), both G = 3 and G = 5 stabilized at very low levels with comparable performance, while G = 10 and G = 15 remained at significantly higher loss values. For the accuracy curves, in the initial stage (first 20 epochs), G = 3 and G = 5 rose at similar rates and outperformed the other groups. In the middle stage (20–100 epochs), both G = 3 and G = 5 rapidly approached 1.0 with minor differences in accuracy, while G = 10 and G = 15 grew relatively slowly. In the later stage (after 100 epochs), G = 3 and G = 5 both stabilized near 1.0 with minimal fluctuation and comparable performance, while G = 10 and G = 15 still had slightly lower accuracy.

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2) Discussion: Overall, G = 3 still exhibits the best comprehensive performance in terms of training efficiency, loss convergence speed, and accuracy stability: on the Yellow River Wetland (autumn) dataset, the advantage of G = 3 is very significant; on the Yellow River Estuary dataset, G = 3 and G = 5 show comparable performance, but G = 3 maintains more efficient convergence. This difference may be attributed to the larger data volume and richer information in the Yellow River Estuary dataset, which allows G = 5 to achieve good performance, yet G = 3 remains a more efficient and stable choice.

The results indicate that simply increasing the number of groups does not necessarily lead to better performance. Under the current experimental settings, G = 3 is an effective and well-balanced choice that balances validity and stability, and this conclusion is verified on both datasets, further demonstrating the robustness and generalization capability of this hyperparameter setting.

Although the selection of G = 3 is empirically justified in this study, its sensitivity should be carefully considered when extending the MPSE mechanism to different datasets or tasks. For scenarios with higher data complexity or more diverse temporal dynamics, a larger or adaptive group number may offer additional flexibility. Future work will explore adaptive strategies for dynamically adjusting G based on dataset characteristics or training convergence behavior, aiming to further improve the model's robustness and generalization across varying application scenarios.

V. Conclusion

This study is guided by the practical requirements of hyperspectral wetland classification and establishes a complete research loop of "data acquisition—problem discovery—model innovation—performance validation." Through a fully autonomous data collection process, we conducted field surveys in the Yellow River Wetlands and collected UAV-based hyperspectral data, constructing a dataset that enables precise transformation from "real-world scenes" to "modeling samples." In-depth analysis of the collected data systematically revealed the core feature of "intraclass heterogeneity" in wetland land cover types. To address the limitations of the traditional Mamba model, we proposed the MPSE mechanism and designed a DP-SS structure to form the integrated Mamba-MPSE model. Experimental results demonstrate that this model outperforms various advanced algorithms based on CNNs, transformers, and the conventional Mamba in wetland classification tasks. The entire research workflow is highly reproducible and provides a paradigmatic reference for synergistic innovation in hyperspectral remote sensing classification, integrating "data—problem—model" [39], [40].

Based on the proposed Mamba-MPSE model and the preliminary results in wetland hyperspectral classification, future research can be further explored in four directions. First, dynamic multipattern adaptive optimization can be advanced. Currently, the number of patterns in MPSE is determined empirically; future work could explore data-driven dynamic pattern selection methods to flexibly adapt to the heterogeneous features of different land cover types. Second, the interpretability and theoretical depth of the mechanism can be enhanced. Future studies could employ feature visualization, gradient attribution, and other techniques to progressively reveal the internal process through which MPSE and DP-SS collaboratively capture intraclass heterogeneity. In addition, the theoretical application of SSMs in hyperspectral data could be further explored, clarifying the mathematical relationship between state evolution and spectral–spatial features of land cover types, thereby laying a solid foundation for hyperspectral sequence modeling. Third, ecological applications can be extended. The current classification model could be applied to wetland ecological assessment tasks, such as vegetation coverage estimation, water boundary extraction, and invasive species monitoring [41]. Combining long-term hyperspectral time series, it is also possible to develop an integrated model for "classification—change detection—trend prediction," providing technical support for dynamic monitoring and degradation risk warning of wetland ecosystems and facilitating the translation of existing technological exploration into actionable ecological management decisions. Fourth, the current Mamba-MPSE model relies solely on hyperspectral data, which limits the integration of complementary information such as elevation and structural features. Recent studies have demonstrated the potential of multisource data fusion for wetland classification. For example, the multisource feature embedding and interaction fusion network (MsFE-IFN) [42] jointly embeds and interacts HSI and LiDAR features to enhance classification, while the instance-wise domain generalization network (IDGnet) [43] addresses cross-scene generalization with HSI and LiDAR data. Incorporating multiple data sources could further enrich intraclass features and improve model generalization, representing a promising direction for future research [44], [45].

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